UNIT 2 · TOPIC 9(B)

Pneumatic Controls

Syllabus topic: Pneumatic and hydraulic controls

The working fluid is compressed **air**. Air is compressible, so pneumatics give lower force density and softer, less precise actuation than hydraulics — but they are clean (no oil contamination of seed/grain), light and fast.

Pneumatics in planting

- **Vacuum (negative-pressure) precision planter:** a rotating metering plate with holes; suction holds ONE seed per hole, and the seed is released where the vacuum is cut off. It is the standard precision metering for maize, cotton, sunflower and vegetables (singulation, minimal seed damage).
- **Air-pressure planters:** positive pressure holds seeds in cells until release.
- **Air seeders/air carts:** a central bulk hopper meters seed into an air stream that distributes it through a manifold and tubes to many furrow openers — wide seeding widths from one hopper.

Other uses

Air-assist/air-blast sprayers (the air stream carries droplets into the canopy), pneumatic conveying of grain, air brakes on heavy trailers, seat suspension.

Comparison

System	Strengths and limits
Hydraulics	High force, stiff, precise; leakage is messy
Pneumatics	Clean, fast, low force; exhaust is free
Electro-mechanical	Precise, efficient, limited force (growing with electrification)

Practice MCQs

Q189. In a vacuum precision planter, seeds are held on the metering plate by:

(a) Positive air pressure (b) Negative pressure (suction)
(c) Glue (d) Magnetism

Q190. An air seeder distributes seed from a central hopper to openers by:

(a) Gravity tubes (b) Augers
(c) An air stream through manifolds (d) Chain conveyors

Q191. Compared with hydraulics, pneumatic systems have:

(a) Higher force density (b) Lower force density but cleaner operation
(c) Same force (d) No actuators

Answers

Q189 — (b) Negative pressure (suction) — *Vacuum plate = singulation by suction (precision planting); air seeder = pneumatic bulk distribution — two different uses of air.*

Q190 — (c) An air stream through manifolds — *Vacuum plate = singulation by suction; air seeder = pneumatic bulk distribution — two different uses of air.*

Q191 — (b) Lower force density but cleaner operation

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.